HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Designed and implemented **thermal assessment methodologies** using **Python scripting** to analyze building envelope performance for 50+ client projects, identifying key areas for energy efficiency improvements.
- Conducted **technical assessments** and calculations for **Building Regulations compliance**, specifically TGD Part L & Part F (NZEB), for new builds and retrofits, ensuring adherence to energy performance standards.
- Prepared detailed **technical documentation**, including drawings and supporting calculations, for **sustainability and energy efficiency projects**, ensuring accuracy and clarity for client and regulatory review.
- Collaborated with architects and engineering teams to review **architectural drawings** and **engineering specifications**, providing input on building performance and sustainability integration.
- Supported the preparation and submission of applications for **SEAI grants** and assisted clients with **building energy upgrade projects**, focusing on practical and cost-effective solutions.
- Assisted in site surveys and inspections to gather data for **building performance testing** and analysis, contributing to the overall assessment of energy efficiency measures.
- Developed **automation scripts in Python** to streamline the data collection and initial analysis phases of thermal assessments, reducing manual processing time by an estimated **30%**.

SKILLS

Building Performance – Sustainability, Energy Efficiency, Building Regulations, TGD Part L, TGD Part F, NZEB, Thermal Assessments, Building Performance Testing

Engineering & Analysis – Mechanical Engineering, Sustainability Engineering, Energy Engineering, Technical Assessments, Calculations, Problem-Solving, Data Analysis

Software & Tools – Python, SQL, Java, Spring Boot, AWS, Docker, Linux, Git, Jenkins, Jira, Confluence, Splunk, Dynatrace, Power BI, Tableau, R, pandas, NumPy, scikit-learn

Documentation & Reporting – Technical Documentation, Drawings, Specifications, Reports, Architectural Drawings, Engineering Specifications

Project Support – SEAI Grants, Building Energy Upgrades, Client Support, Stakeholder Collaboration, Site Surveys, Inspections

PROJECTS

Building Envelope Performance Analysis Tool

- Developed a **Python-based tool** to automate the analysis of building envelope thermal performance data, integrating calculations for U-values and R-values.
- Designed the tool to process input data from client surveys and architectural drawings, generating initial reports on energy loss pathways.
- Implemented basic **data visualization** features using Matplotlib to highlight key areas for improvement, aiding in client understanding of thermal bridging and insulation deficits.

NZEB Compliance Reporting Framework

- Created a framework using **Java Spring Boot** and **PostgreSQL** to manage and track compliance documentation for NZEB (Nearly Zero-Energy Building) requirements.
- Engineered REST APIs to facilitate data exchange between the reporting framework and client project management systems, enabling streamlined document submission.
- Designed database schemas and wrote SQL queries for efficient retrieval and aggregation of compliance data across multiple projects, supporting regulatory audits.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)